



Exploring the Correlation between Pediatric Healthcare Outcomes and Socioeconomic variables using KID's Inpatient Data

**BSAN 710: Business Analytics Project
Team 1**

Executive Summary



Effects of Socio-Economic Factors on Alcohol and Substance Abuse in children

- Females are charged more in alcohol abuse treatments on average than males.
- Charges for Medicaid are higher compared to private .
- Private owned Investor hospitals are charging lesser than non-profits.

Subsidized Care for female child



Correlation of Socio-Economic Factors on Neonatal Deaths

- Being underweight followed by Respiratory Issues are the most prevalent issues for neonatal deaths.
- Asian and Pacifics paying higher than others even though they are having less population.
- Patients from the median income bracket of 61,000-81,999 have the highest average charges in south.

Stricter tobacco policies for pregnant women.



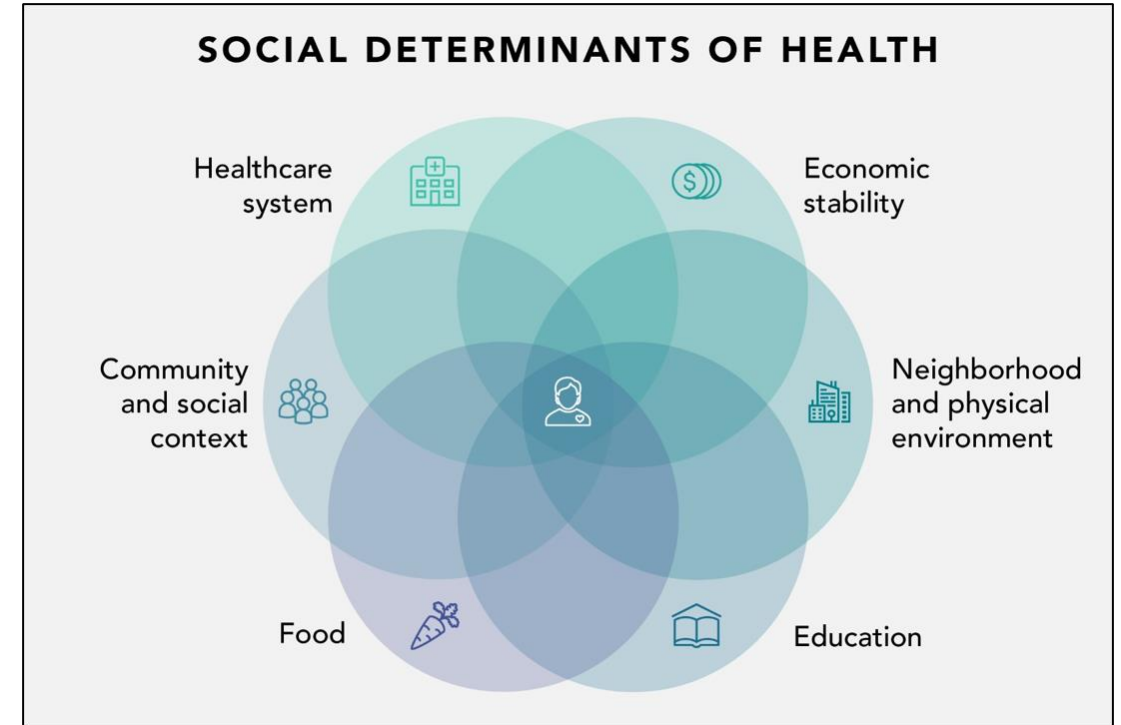
Streamlining medical billing processes to save ~ **\$500 Billion** and fostering public-private insurance collaboration to increase Medicaid acceptance among providers, thereby improving neonatal and substance abuse outcomes.

Business Questions

1. Effects of Socio-Economic Factors on Alcohol and Substance Abuse in children
2. Correlation of Socio-Economic Factors on Neonatal Deaths

Socio Economic Factors Include:

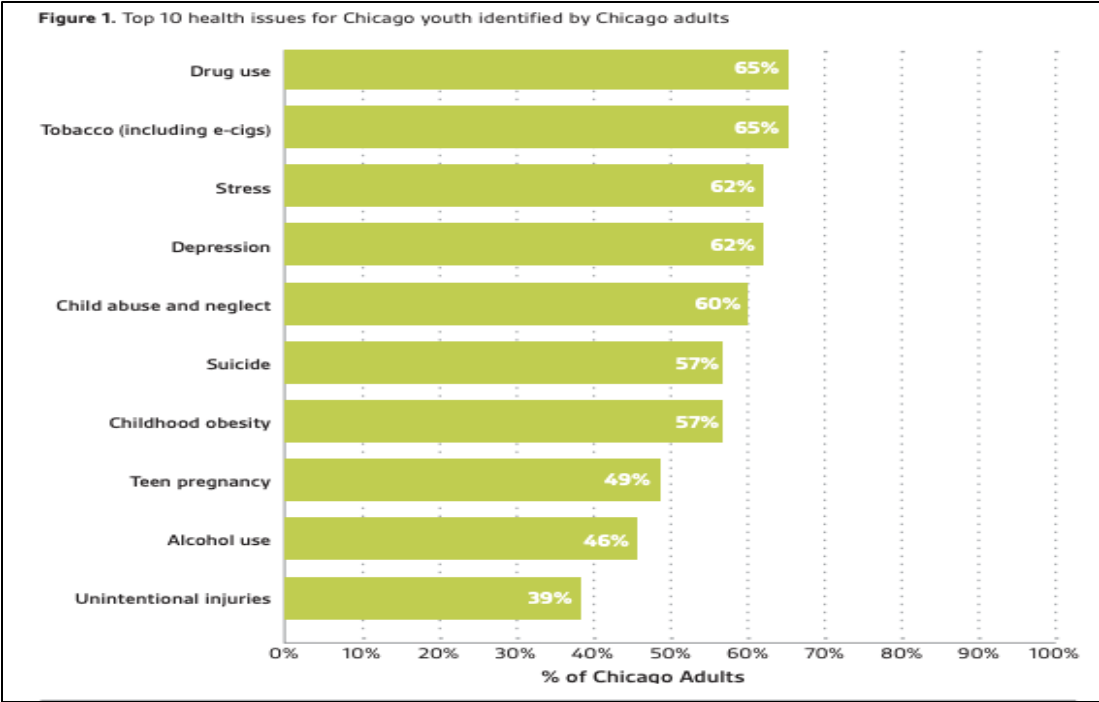
1. Race
2. Region
3. Income
4. Gender
5. Age



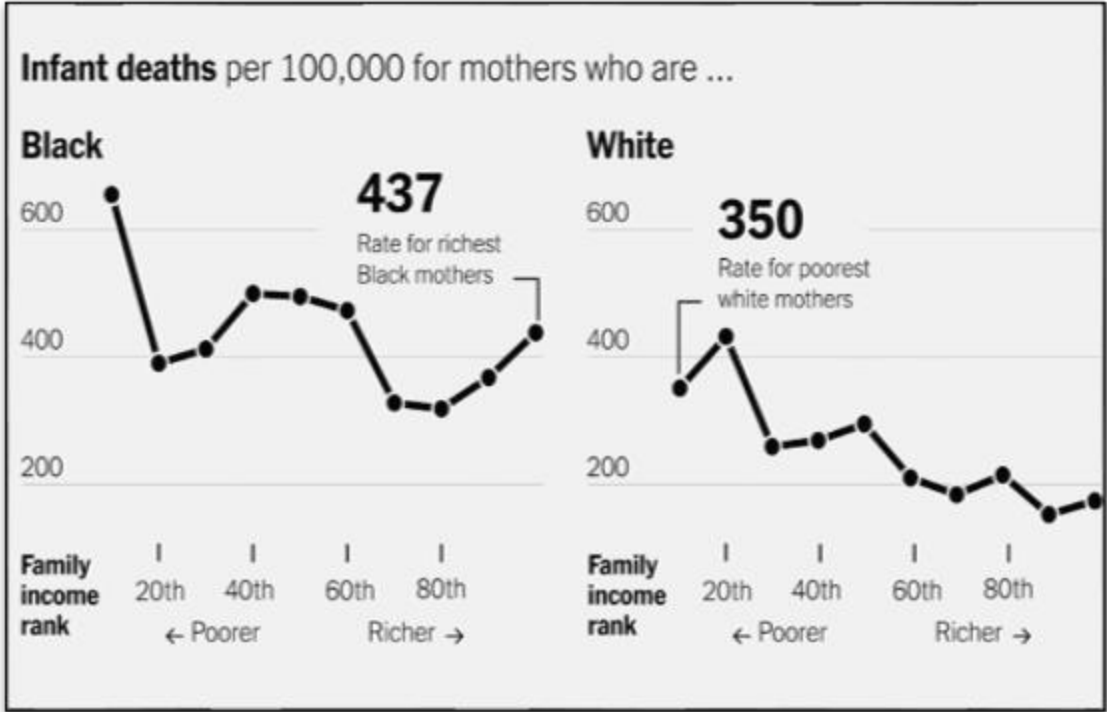
<https://healthinfo.montana.edu/cheg/health-equity/index.html>

Caution: The following slides may contain sensitive content. Viewer discretion is advised.

Motivation



<https://www.luriechildrens.org/en/voices-of-child-health-in-chicago/top-10-youth-non-covid-19-health-problems/>



<https://www.nytimes.com/interactive/2023/02/12/upshot/child-maternal-mortality-rich-poor.html>

Caution: The following slides may contain sensitive content. Viewer discretion is advised.

Chi Squared, Kruskal Wallis, and Dunn's Test

CHI SQUARED TEST

- Chi Squared Test used to identify relation between Socio Economic Factors on research variable.

KRUSKAL WALLIS AND DUNN'S TEST

- Kruskal Wallis Test used to test difference in charges across Regions and Races.
- Dunn's Test is a post-hoc procedure to identify which specific groups differ.



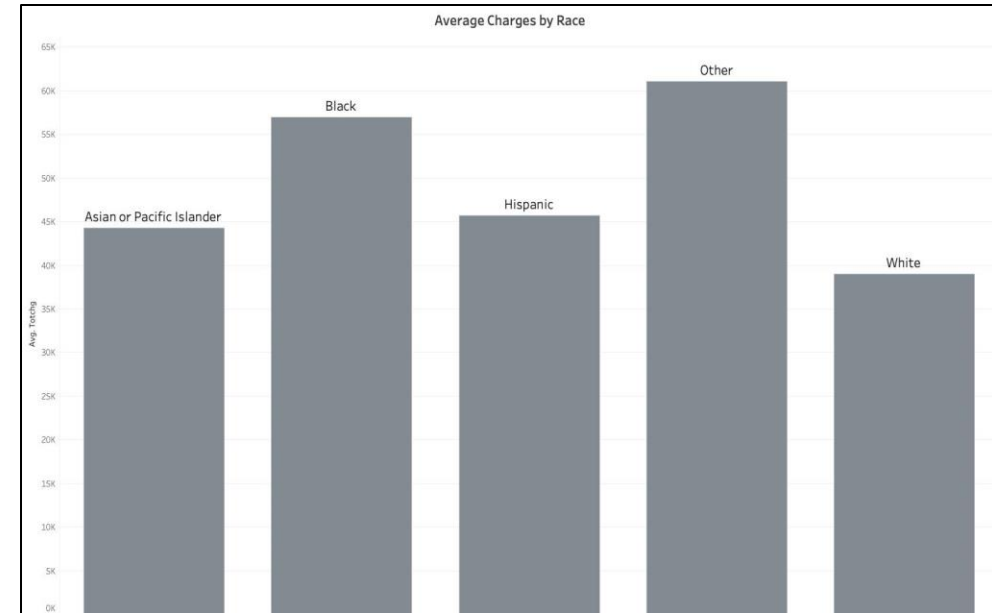
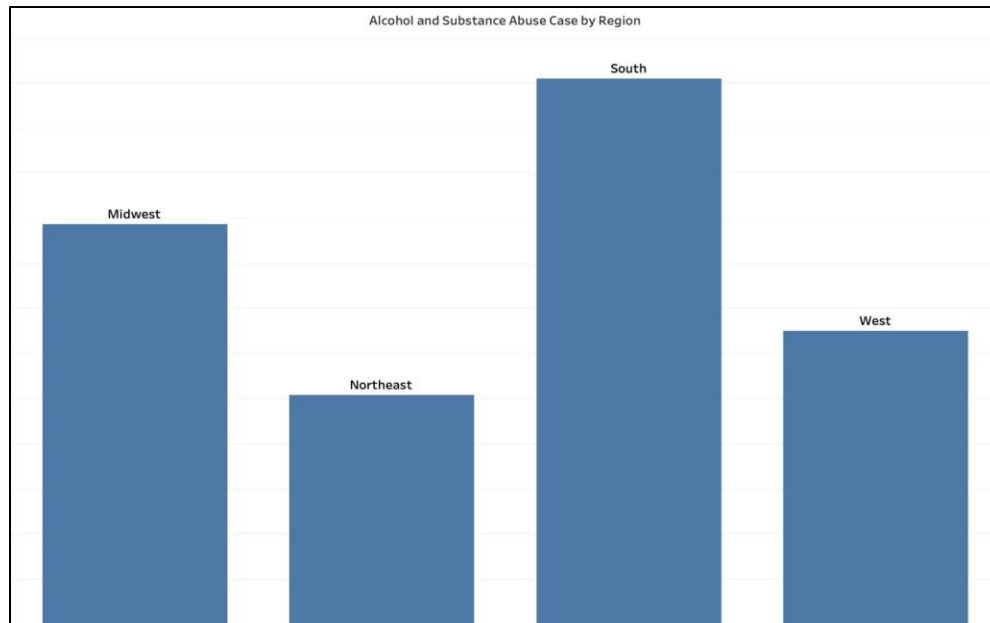
We reject Null Hypothesis for both tests, indicating -

- Significant relation between each socioeconomic factor and target variables.
- significant difference in the average total charges between the different hospital regions and races.

Hypothesis Output for Alcohol & Substance Abuse

Dunn's Test Results for Alcohol & Substance Abuse

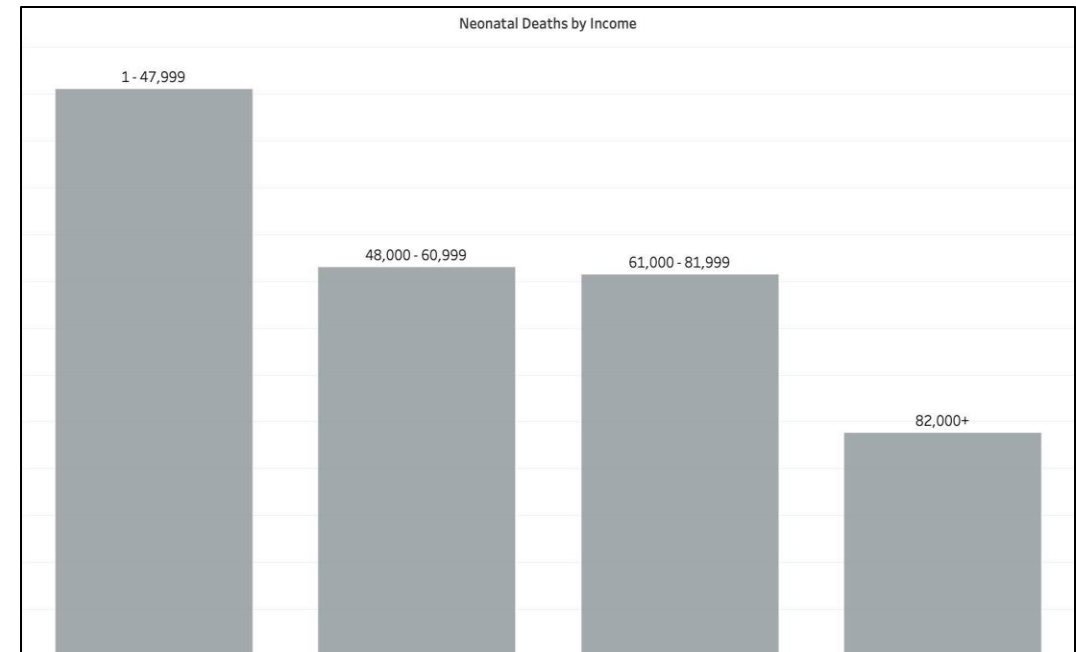
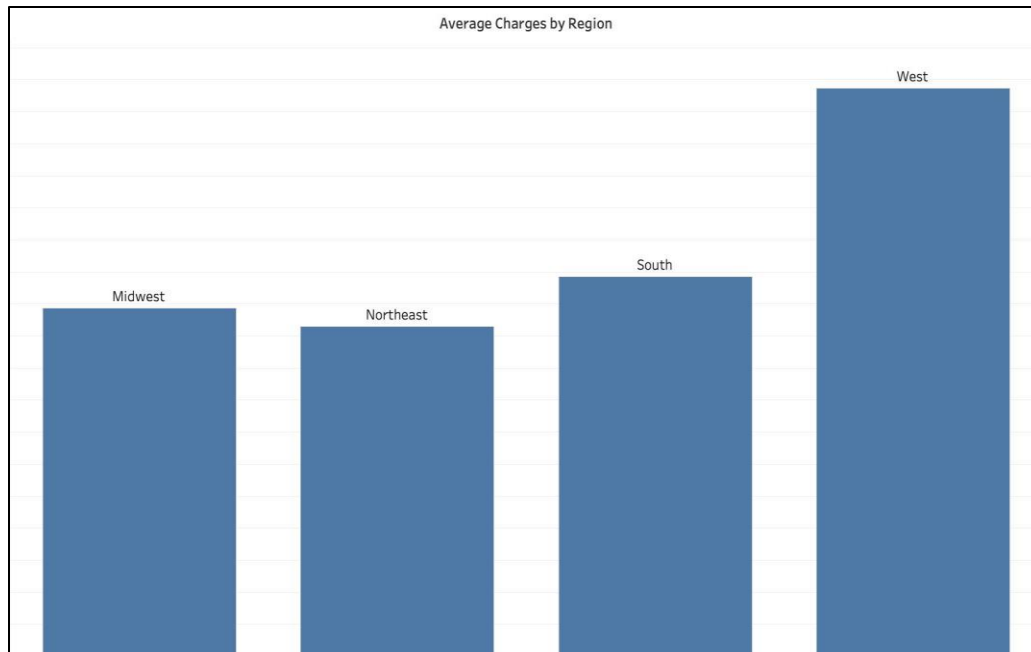
- **Northeast vs. West:** No significant differences.
- Average Charges different for all other regional combinations.
- **Asian or Pacific Islander vs Hispanic:** No significant difference in average charges.
- **Asian or Pacific Islander vs Black:** No significant difference in average charges.
- Average Charges different for all other racial combinations.



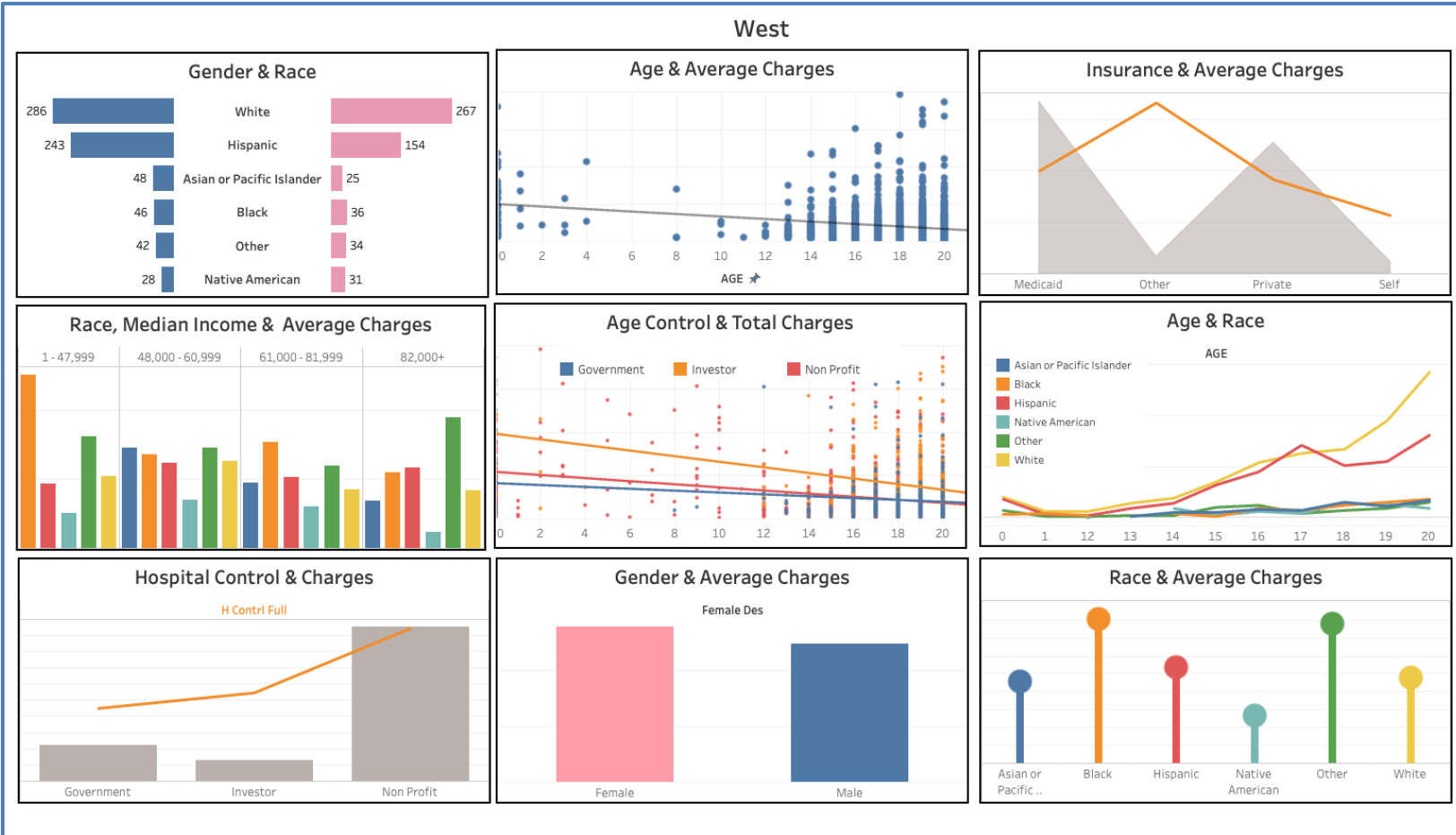
Hypothesis Output for Neonatal Death

Dunn's Test Results for Neonatal Death

- **Northeast vs. South:** No significant differences in average charges.
- **Northeast vs. Midwest:** No significant differences in average charges.
- Average Charges different for all other regional combinations.
- **White vs. Other Races:** No significant differences in average charges.
- **Asian or Pacific Islander vs. Other Races:** No significant differences in average charges.
- Average Charges different for all other racial combinations.

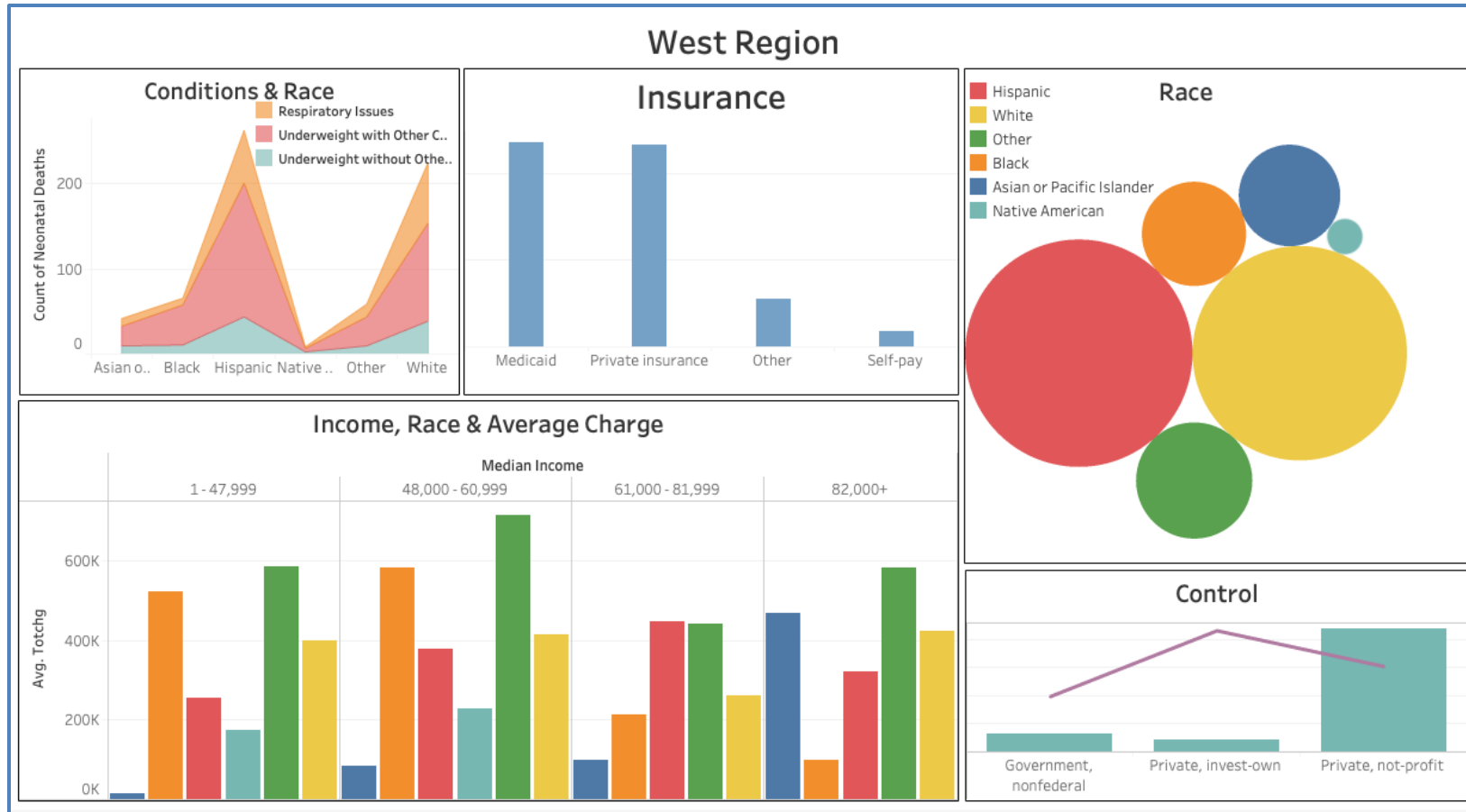


Insights from Regional Analysis: Alcohol & Substance Abuse



- Average Cost decreasing by Age.
- Females on average are charged higher for services compared to Men.
- In lower income brackets, Blacks and Asian & Pacific Islanders are being charged the most compared to other races.
- Medicaid & Private Insurance cost almost similar.
- 'Private Non-profit' Hospitals cost more than 'Private Investor Owned' Hospitals.

Insights from Regional Analysis: Neonatal Deaths



- Whites and Hispanics show higher instances of Respiratory Issues and Underweight Infants.
- Almost similar count of deaths for those who opt for Medicaid vs Private Insurance.
- Higher Death neonatal death count for Hispanics and White as compared to other races.
- In lower income brackets, Blacks are charged higher on average compared to other races.

Streamlining Billing and Insurance Related Activities (BIR)

- BIR constitutes- 12.3% on private insurance; and 3.5% on public program in 2010
- Estimated BIR costs \$496 billion for 2019
- 2019 estimates- \$158 billion in overhead for private insurance; \$56 billion for administration of public insurance programs
- Moreover, 19% less hospitals prefer Medicaid to Private Insurance due to Billing Related Issues.

Streamlining Billing Process Benefits:

- Could reduce the administrative costs to approximately 12% of revenues from 25%.
- Will address the around \$504 billion excess cost in \$1.1 trillion currently spent on health care administration

Administrative costs in the U.S. health care system

Estimated costs related to billing and insurance, 2019

Category	Annual cost in billions
Providers (physicians, hospitals, and others)	\$282
Private insurers	\$158
Public programs	\$56
Total	\$496

Source: Authors' calculations based on National Academy of Medicine, "Excess Administrative Costs," in Pierre Yong, ed., The Healthcare Imperative: Lowering Costs and Improving Outcomes: Workshop Series Summary (Washington: National Academies Press, 2011); Centers for Medicare and Medicaid Services, "National Health Expenditure Data," available at <https://www.cms.gov/research-statistics-data-and-systems/statistics-trends-and-reports/nationalhealthexpenddata/nhe-fact-sheet.html> (last accessed January 2019).



NICU related to Neonatal Health Care Cost

- Maternal smoking is related to nearly a 20% increase in the relative risk of NICU admission and longer stays for infants in the NICU.
 - Smoking elevates infant length of stay by 1.1%, with NICU nights costing \$2496, compared to the \$1796 cost in a regular nursery.
 - Maternal smoking contributes a 2.2% Smoking Attributable Fraction, resulting in an additional \$700 per birth in neonatal costs, totaling nearly \$367.
- **Recommendation:** Advocate for targeted smoking cessation in African, American, and White communities and stricter tobacco policies to assist pregnant women in quitting.

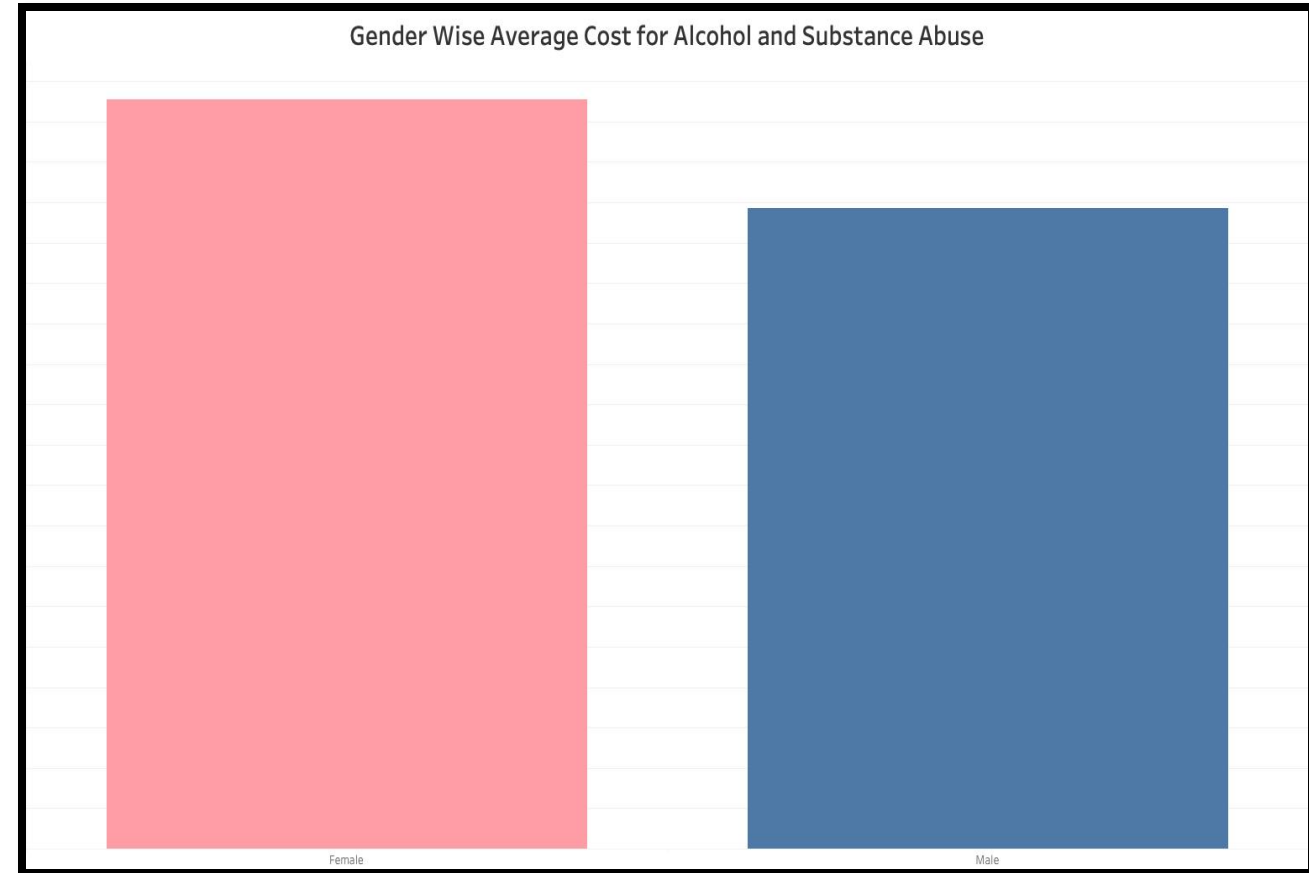
Percentage of mothers who smoked during pregnancy, 2020		
Non-Hispanic Black	Non-Hispanic White	Non-Hispanic Black /Non-Hispanic White Ratio
4.5	8.1	0.6

Socio-Economic Status, Alcohol and Substance Use in Children

- **Gender Disparity:** Females charged more than males, especially in the Midwest and Northeast.

Recommendation:

- **Subsidized Care for Women:** Programs similar to those in some European countries, offering additional subsidies for women, especially with dependents, for equal treatment access.



Socio-Economic Status (SES), Alcohol & Substance Use in Children

Alcohol and Substance Abuse with respect to Income quartiles

Smoking:

- 26.7% in the bottom income quartile smoke, compared to 16.5% in the top.

Alcohol and Marijuana Use:

- 75.8% in top income quartile drink alcohol vs. 51.0% in bottom.
- 28.8% in top quartile engage in heavy episodic drinking.
- 39.1% in top income quartile use marijuana.

Recommendations :

- **Tailored Prevention Programs:** Focus on smoking in lower SES areas and alcohol/marijuana use in higher SES regions.
- **Parental Engagement:** Involve parents, especially in higher SES communities, to mitigate substance use.

Source:- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3410945/>

School-based drug abuse prevention programs

❖ Botvin's LifeSkills training

- Based on more than 40 years of peer reviewed scientific research. LST is effective universally, across diverse populations and settings.



Cost Benefit Analysis of Botvin's LifeSkill Training

- Overall US under 18 population- 73 Million
- Philadelphia students under 18 population- 118,053 (0.16% of US under 18 population)
- Streamlining billing process will save around \$504 Billion
- Botvin program cost per student- \$235
- Cost to enroll Philadelphia under 18 children- **\$28 Million**
- This is **less than 1%** of the savings that could be used to enroll students in Philadelphia
- There is another program called project Towards No Drug Abuse, which is also proved to be successful.

School-based drug abuse prevention programs

❖ Project TND (Towards No Drug Abuse)

- Selected (for members of at-risk groups)
- seven research trials and over 8,600 youth participants, to date, across ethnicities

Evidence of Effectiveness

- 25% decrease in hard drug use (one-year follow-up).
- 7-12% reduction in alcohol use for prior users.
- 27% drop in cigarette use (one year), 50% (two years).
- 22% decrease in marijuana use (one year), significant reduction in hard drug use (two years)

References

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Thank You.